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The Impact of a Minimum Wage Increase on Hourly Wages and Total Income: A Case Study of the Young Adult Labor Markets in Nebraska and Oklahoma

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In 2015, Nebraska increased its minimum wage from \$8 to \$9 an hour through a ballot initiative. To evaluate the impact of this minimum wage increase on hourly wage and total income of young workers, we leverage an analysis of the labor force aged 16-25 in Nebraska and Oklahoma - a state where the minimum wage remained stable during the same period. We employ a differences-in-differences approach to compare trends in Nebraska and Oklahoma before and after the minimum wage increase, focusing on hourly wage and total income as outcomes. Our analysis finds that the increase in minimum wage did not have a statistically significant effect on hourly wage or total income for workers aged 16-25.

I. Introduction

According to the foundational microeconomic theory of supply and demand, in a perfectly competitive labor market, an increase in worker's wages should lead to a decrease in employment as an employer's demand for labor decreases. However, in reality where there are many other influencing factors, the impact of an increase in worker's wages is more ambiguous. In 1994, Nobel Prize-winning economists David Card and Alan Krueger published their influential paper investigating the causal effect of an increase in minimum wage on fast-food restaurant employment. They utilized data from the natural experiment that occurred in 1992, when New Jersey increased its minimum wage from \$4.25 to \$5.05 an hour, while Pennsylvania did not. Under the parallel trends assumption, the economists concluded that the labor market of Pennsylvania served as an appropriate counterfactual for the labor market of New Jersey due to

their proximity in both location and economic trends. In doing so, they found that New Jersey's employment in fast food restaurants increased by 13 percent compared to stores in Pennsylvania, which is counter to the expectations of the supply-and-demand model. Using this framework, we sought to determine the causal effect of an increase in minimum wage on hourly wage and total income for workers aged 16-25.

A. Research Questions

1. Does an increase in the minimum wage definitively increase the hourly wage of young workers aged 16-25?
2. How does an increase in the minimum wage affect the total income of these young workers?

B. Motivation for Examining Nebraska and Oklahoma

From 2015 to 2016, Nebraska increased their minimum hourly wage from \$8 to \$9 while the legally-mandated minimum hourly wage remained stable in Oklahoma. To determine the effect of this change, we would need to consider what would have happened in Nebraska had the change not occurred, which is an unobservable counterfactual. As a result, we decided to utilize Oklahoma as the comparison state for Nebraska for multiple reasons. First, at the same time as Nebraska's minimum wage increase, Oklahoma adhered to national minimum wage laws and remained at \$7.25 an hour. Oklahoma's minimum wage did not change throughout the years we examine (2011-2019). Further, as neighboring states that share geographic characteristics; political trends; and a focus on the agriculture, manufacturing, and energy industries, we inferred that their labor markets would be statistically similar. Additionally, both states have a

comparable rural-urban demographic mix. Finally, Nebraska and Oklahoma have similar size economies, where Nebraska's total annual gross domestic product (GDP) per capita was \$58,664 in 2013 while Oklahoma's was \$47,288 in the same year (Legislative Research Office).

C. Motivation for Examining Workers Aged 16-25

This paper specifically focuses on the causal effects of Nebraska's increased minimum wage on workers aged 16-25. We seek to investigate workers who are most sensitive to changes in the minimum wage, which is likely individuals employed in minimum wage roles. In this age range, workers are less likely to possess prior work experience or specialized skills, which means they fill a higher proportion of minimum wage positions with relatively low barriers to entry compared to those of salaried jobs. Although they might work at the minimum wage or earn lower total incomes compared to older workers, this age range is crucial for both these Great Plains states. First, workers between the ages of 16-25 are active contributors to state economies through consumer spending, especially on housing, entertainment, and transportation. Additionally, this age range represents a crucial consideration of politicians at the local and state level due to growing voter mobilization among young people. Finally, the workers of this age bracket play a pivotal role in determining the demographics and long-term population growth, as this is the period of transition in which individuals choose to remain in-state or migrate elsewhere in search of better job opportunities or higher wages.

D. Motivation for Examining Hourly Wage and Total Income

One argument made for the increase of minimum wages is that it would move many minimum wage individuals and families out of poverty by providing them with an income that is

appropriate for increased costs of living. Contrastingly, it is also argued that the increase in minimum wage would increase operating costs for businesses, causing their costs of service to rise and thus canceling out any effects on improving poverty rates and total income (Storer et al., 2021). Similar to this dichotomy, the relationship between minimum wage increase and total income is also ambiguous. On one hand, an increase in minimum wage would increase total income by nature of earning more per hour worked. Contrastingly, it could negatively impact total income in young workers by raising unemployment rates and costs of living. For these reasons, using population data from the years 2011-2019, we chose to focus on the causal effect of an increase in minimum wage upon both hourly wage and individual total income, controlling for individual characteristics such as race, sex, and school attendance. Through statistical analysis and the differences in differences method, we found the following:

- a. Using the natural log of hourly wage, we found that only in 2018 was there a significant difference in the natural log of hourly wage between Nebraska and Oklahoma, with Oklahoma having a slightly higher value than Nebraska.
- b. Using the natural log of total income variable, we determined that the mean total income for young workers in both Nebraska and Oklahoma rose after the treatment year of 2015, with Nebraska's value being significantly greater than Oklahoma.

II. Data

A. Source of Data and Construction of Sample

We drew our dataset from the American Community Survey (ACS) available on ipums.org, which provides census and survey data from the United States and its territories. After downloading the sample of all respondents from Nebraska and Oklahoma during the 2011-2019 period, we were able to filter and clean the data to narrow our focus on a specific group of individuals. In Stata, we removed missing values from the data set, then filtered for ages 16-25, positive income, and the non-institutionalized population, leaving a restricted analysis of non-institutionalized individuals who report their educational history and total income in Nebraska and Oklahoma from the years 2011-2019 (four years before and after Nebraska's minimum wage change in 2015).

B. Variables

In total, we downloaded 26 variables from IPUMS, including our main 3 controls of race, sex, and school attendance. The 'statefip' variable reports which state the worker resides in using the alphabetical Federal Information Processing Standards. The two outcome variables of interest are each worker's wages and total income, but only one was available directly from the IPUMS website. Specifically, 'inctot' was provided from the IPUMS site, and discloses an individual's total pre-tax personal income or losses from all sources for the previous year.

C. Constructed Variables

In order to implement our differences-in-differences regressions, we construct several variables within our data set. First, we construct an indicator variable to reflect whether a sample

is in our treatment state or control state. We thus generate the variable ‘treatment’ for those observations in which the ‘statefip’ variable is equal to 40 (the IPUMS FIP code for Nebraska). Next, we construct an indicator variable to identify whether a sample is drawn from before or after the minimum wage increase. Given Nebraska implemented the minimum wage increase in 2015, we generate the variable ‘post,’ taking the value of one if the sample’s ‘year’ variable is 2015 or later and otherwise taking the value of zero. Since the interaction term’s coefficient is what actually represents the causal estimate in a differences-in-differences analysis, we generate the interaction variable ‘post_treatment’ in our dataset by multiplying the ‘post’ indicator by the ‘post’ indicator.

Since IPUMS does not provide a variable that directly represents our second key outcome variable of hourly wage, we calculated the variable ‘ln_hrly_wage’ using three variables that it did provide: ‘incwage,’ ‘wkswork2,’ and ‘uhrswork.’ The variable ‘incwage’ consists of an individual’s annual income from wages and salaries, ‘wkswork2’ represents a range of respondent’s weeks worked in a year, and ‘uhrswork’ represents an individual’s usual hours worked in a week. As the number of weeks worked for each respondent was only available in intervals (1-13 weeks, 14-26 weeks, etc), we calculated the midpoint of each interval. We then used this midpoint to create an ‘hrly_wage’ variable by dividing ‘incwage’ by the midpoint of ‘wkswork2’ multiplied by ‘uhrswork.’ Since we are interested in the percent change in the hourly wage instead of the dollar value, we took the natural log of this variable to create ‘ln_hrly_wage.’

Additionally, following our logic in taking the natural log of our constructed ‘hrly_wage,’ we logged the ‘inctot’ variable that was constructed by IPUMS in order to interpret the coefficients produced as comparable percent changes.

Finally, to control for confounding factors and reduce omitted variables bias, we also created several dummy variables to convert categorical values for race, sex, and school attendance into quantitative measures. We created dummy variables for many racial categories using their codes as provided in IPUMS. For example, we generated ‘race_black’ for those observations in which ‘race’ was equal to 2 (the IPUMS code for a black respondent). We used this same process to create 7 other racial dummy variables. Next, we created the ‘female’ variable, taking the value of one if the sample’s ‘sex’ code is equal to two (the IPUMS code for female) and otherwise zero. Finally, we created the control variable ‘in_school’ to take the value of one when a sample’s ‘school’ variable was equal to 2 (the IPUMS code for attending school) and otherwise zero. This control is necessary to narrow our focus toward those workers who would be most impacted by the minimum wage increase.

III. Empirical Strategy

A. Regression Approach

We utilized the differences-in-differences approach to test how the minimum wage affected hourly wage and total income for our chosen subgroup. We did this by analyzing the change in the calculated hourly wage for workers aged 16-25 in Nebraska compared to Oklahoma, and also the change in total income for this age bracket in Nebraska versus Oklahoma. We control for state fixed effects (which are characteristics of Nebraska and Oklahoma that might differentiate the states from each other but remain fixed over time) by absorbing the ‘statefip’ variable from IPUMS in our regression model. We also included dummy variables for each year in our period of analysis (yeardum*) to account for broader

macroeconomic trends or shocks that would have impacted both Nebraska and Oklahoma in the same year. Including these additional controls allows our regression to isolate the effect of the minimum wage increase on hourly wage and total income instead of being biased by omitted variables.

We estimate the following regression models:

$$(1) \ln_hrly_wage_i = \beta_0 + \beta_1 treatment + \beta_2 Post + \beta_3(treatment \times Post) + u_i$$

$$(2) \ln_hrly_wage_i = \beta_0 + \beta_1 treatment + \beta_2 Post + \beta_3(treatment \times Post) + \beta_4 in_school + \beta_5 native_american + \beta_6 other_asian_pacific + \beta_7 race_other + \beta_8 race_two_major_races + \beta_9 race_three_or_more_races$$

$$(3) \ln_inctot_i = \beta_0 + \beta_1 treatment + \beta_2 Post + \beta_3(treatment \times Post) + u_i$$

$$(4) \ln_inctot_i = \beta_0 + \beta_1 treatment + \beta_2 Post + \beta_3(treatment \times Post) + \beta_4 in_school + \beta_5 native_american + \beta_6 other_asian_pacific + \beta_7 race_other + \beta_8 race_two_major_races + \beta_9 race_three_or_more_races$$

where ‘Post’ is a dummy variable for observations after Nebraska’s minimum wage change in 2015 and ‘treatment’ is a dummy variable for Nebraska. Because the balance test displays a significant difference in the changes in share of the population that is in school (shown in Figure 1 below), we run regressions including a control for ‘in_school’, ‘native_american’, ‘other_asian_pacific,’ ‘race_other,’ ‘race_two_major_races,’ and ‘race_three_or_more_races.’

We also estimate the following regression models using state fixed effects:

$$(5) \ln_hrly_wage_{ist} = \beta_0 + \beta_1 treatment + \beta_2 Post + \beta_3(treatment \times Post) + \alpha_s + \delta_t + u_{ist}$$

$$(6) \ln_hrly_wage_{ist} = \beta_0 + \beta_3(treatment \times Post) + \beta_4 in_school + \beta_5 native_american + \beta_6 other_asian_pacific + \beta_7 race_other + \beta_8 race_two_major_races + \beta_9 race_three_or_more_races + \alpha_s + \delta_t + u_{ist}$$

$$(7) \ln_inctot_{ist} = \beta_0 + \beta_3(\text{treatment} \times \text{Post}) + \alpha_s + \delta_t + u_{ist}$$

$$(8) \ln_inctot_{ist} = \beta_0 + \beta_3(\text{treatment} \times \text{Post}) + \beta_4 \ln_school + \beta_5 \text{native_american} + \\ \beta_6 \text{other_asian_pacific} + \beta_7 \text{race_other} + \beta_8 \text{race_two_major_races} + \\ \beta_9 \text{race_three_or_more_races} + \alpha_s + \delta_t + u_{ist}$$

where α_s represents state fixed effects, δ_t represents year fixed effects, and $\ln_hrly_wage_{ict}$ represents the log wage of person i in state s in year t . We assume that, absent Nebraska's change in minimum wage, changes in hourly wages and total income would have been the same in Nebraska and Oklahoma.

B. Necessary Assumptions

For our difference-in-differences analysis to yield causal effects of the 2015 minimum wage increase in Nebraska on hourly wage and total income for young workers, we rely on the parallel trends assumption. This assumption implies that absent the minimum wage increase, changes in hourly wage and total income in Nebraska would have mirrored the changes in Oklahoma during the same period. While we cannot directly observe the counterfactual trend in Nebraska, we perform several tests to evaluate the plausibility of this assumption.

To assess whether pre-trends in hourly wage and total income were parallel between Nebraska and Oklahoma, we graph the mean natural log of hourly wage (' $\ln_hrly_wage$ ') and total income (' $\ln_inctot$ ') over time for both states (Figures 4 and 5). The visual evidence demonstrates similar upward trajectories in both variables before 2015, the year the policy was implemented. The overlapping confidence intervals for both states in the pre-treatment period suggest no statistically significant differences in the trends prior to the minimum wage increase. This supports the validity of the parallel trends assumption for our primary outcome variables.

In addition to evaluating pre-trends, we conducted a balance test (Figure 1) to compare demographic characteristics of Nebraska and Oklahoma prior to 2015. Significant differences in school attendance and some racial categories are evident, as indicated by the coefficients for ‘in_school’ and certain racial dummy variables. To account for these differences and reduce potential bias, we include these controls in our regressions. The balance test supports that Nebraska and Oklahoma are comparable in key respects, though some differences necessitate the inclusion of demographic controls.

While pre-trend comparability provides initial support for our analysis, we must also assume that no state-specific shocks or policy changes, other than the minimum wage increase, differentially affected young workers in Nebraska or Oklahoma during the post-treatment period. For example, macroeconomic shocks such as regional economic changes, natural disasters, or industry-specific declines could violate the parallel trends assumption. By including state and year fixed effects in our regressions, we aim to isolate the treatment effect and account for these potential confounders.

While these tests and controls provide strong evidence for the validity of our assumptions, it is important to recognize the limitations inherent in quasi-experimental research. Any unobserved factors that differentially affect Nebraska and Oklahoma during the study period could bias our estimates. Nevertheless, the robustness of our findings across specifications suggests that the results are not driven by violations of the parallel trends assumption.

IV. Results

A. Findings

The findings of our difference-in-differences approach to the causal effect of the minimum wage increase in Nebraska on hourly wage and total income, using Oklahoma as a control group, are summarized in Figure 3.1 and 3.2. Figure 3.1 displays regressions of logged hourly wage ('ln_hrly_wage') on the interaction term ('Post x Treatment') under different model specifications. Furthermore, Figure 3.2 displays the regression results of logged total income ('ln_inctot') under different model specifications. In Figure 3.1, column (1) captures the regression of the natural log of hourly wage on the treatment-post interaction without additional controls. Columns (2), (3), and (4) show the regression of the natural log of hourly wage on the post-treatment interaction and the statistically significant controls of race and school attendance (as shown in the balance test). Similarly, column (1) in Figure 3.2 displays the regression of the natural log of total income on the post-treatment interaction, with columns (2), (3), and (4) including controls for race and school attendance. In both Figures 3.1 and 3.2, columns (5) - (8) capture state and time fixed effects.

In Figure 3.1, regression coefficients in columns (5) and (8) represent the additional percent change in 'ln_hrly_wage' from before 2015 to after in Nebraska compared to Oklahoma, with state and time fixed effects in addition to the constructed control variables. Regression coefficients in columns (5) and (8) represent the additional percent change in hourly wage from before 2015 to after 2015 in Nebraska compared to Oklahoma, with only the race and school attendance controls and no fixed effects. From column (1) in Figure 3.1, we see that with no controls for fixed state and time effects or other individual characteristics, Nebraska experiences a 2.3 percent additional decrease in 'ln_hrly_wage' from before 2015 to after relative to the change in Oklahoma, which is not statistically significant at the 10 percent level. We see that in column (4), which controls for race and school attendance, this change shrinks to a 2.0 percent

decrease yet this change is insignificant at a 1%, 5%, and 10% significance level. However, from column (5) in Figure 3.1, we see that when controlling for fixed state and time effects, the `post_treatment` interaction variable demonstrates that relative to the change in Oklahoma, Nebraska's hourly wage decreased by 2.3% more than the decrease in Oklahoma from before 2015 to after 2015, which is significant at the 5 percent level. With additional controls for race and school attendance, column (8) denotes this change reduces to 2 percent, which is significant at the 1 percent level.

Furthermore in Figure 3.2, regression coefficients in columns (5) and (8) represent the additional percent change in `ln_inctot` from before 2015 to after 2015 in Nebraska compared to Oklahoma, with state and time fixed effects as well as controls. From column (1) in Figure 3.2, we see that with no controls for fixed state and time effects or other individual control characteristics. The `post_treatment` variable explains that Nebraska experiences a 0.7 percent additional decrease in total income from the period before 2015 to the period after relative to the change in Oklahoma, which is not statistically significant at the 10 percent level. We see that in column (4), which controls for race and school attendance, this change in the `post_treatment` variable switches to a 0.6 percent increase, however, this is still statistically insignificant as well. In column (5) in Figure 3.2, we control for fixed state and time effects, Nebraska experiences a 0.7 percent additional decrease in total income from before 2015 to after 2015 relative to the change in Oklahoma, which is significant at the 1 percent level. With additional controls for race and school attendance in column (8), this change reduces to 0.6 percent, which is significant at the 10 percent level but not the 5 percent level.

B. Discussion

Our results suggest that the 2015 minimum wage increase had modest effects on the hourly wage and total income of young workers aged 16-25 in Nebraska. Intuitively, an increase in the minimum wage would be expected to raise or at least maintain hourly wages. However, our findings show that hourly wages in Nebraska decreased post-policy implementation. When controlling for state and time fixed effects alongside demographic controls, this decrease was statistically significant at the 1% level. This indicates that the minimum wage increase likely played a role in mitigating further wage decreases but did not result in the expected increases for young workers in Nebraska.

The regressions without fixed effects fail to produce significant results, which can be attributed to several factors. First, the absence of state and time fixed effects leaves the model susceptible to unaccounted heterogeneity between Nebraska and Oklahoma. Differences in labor market structures, economic growth rates, and state-specific policies unrelated to the minimum wage increase may bias the estimates. These unobserved factors, not explicitly accounted for, contribute to noisy and insignificant estimates. Second, the lack of fixed effects hinders the model's ability to fully satisfy the parallel trends assumption, which is central to the Difference-in-Differences (DiD) methodology. While Figures 4 and 5 illustrate largely parallel trends in wages and income between Nebraska and Oklahoma pre-treatment, subtle deviations in these trends could mask the true treatment effect in the absence of explicit controls. Finally, the exclusion of fixed effects introduces omitted variable bias, as important factors such as state-specific economic shocks and federal policy changes are not accounted for. This likely inflates the variance of the error term, resulting in larger standard errors and reducing the precision of the estimates.

In contrast, regressions incorporating state and time fixed effects yield statistically significant results, highlighting the importance of these controls in isolating the treatment effect. State fixed effects account for time-invariant characteristics unique to Nebraska and Oklahoma, such as baseline economic structures, demographics, and institutional frameworks. This ensures that the treatment effect is derived solely from within-state changes over time, rather than cross-state comparisons. Time fixed effects capture nationwide shocks or trends that impact both states equally, such as federal policy changes or macroeconomic conditions, further refining the estimates. By addressing unobserved heterogeneity, fixed effects reduce the variance of the error term, yielding smaller standard errors and more precise estimates. This improvement explains the shift from insignificant to significant results between the non-fixed-effects and fixed-effects regressions.

Interestingly, the inclusion of fixed effects also raises concerns regarding multicollinearity, which may contribute to the observed near-zero standard errors in some regressions. Multicollinearity arises when fixed effects are included alongside other highly correlated variables, such as state and time dummies. While this does not invalidate the coefficients, it does necessitate cautious interpretation of the standard errors and the precision of the estimates.

The results for hourly wages, presented in Figure 3.1, offer a clearer perspective on these dynamics. Without fixed effects, the regression suggests a 2.3% additional decrease in Nebraska's hourly wages relative to Oklahoma following the minimum wage increase. Adding controls for race and school attendance slightly reduces this estimate to a 2.0% decrease. However, both results are statistically insignificant, reflecting the limitations of unaccounted heterogeneity and omitted variables. With fixed effects, the treatment effect remains negative but

becomes statistically significant. Controlling for state and time fixed effects, the hourly wage decreased by 2.3% in Nebraska relative to Oklahoma, significant at the 5% level. Adding demographic controls reduces this estimate to 2.0%, significant at the 1% level. These findings suggest that the minimum wage increase had a modest yet statistically meaningful impact on wages when broader contextual factors were accounted for.

The results for total income, as shown in Figure 3.2, follow a similar pattern. Without fixed effects, the regressions show small and insignificant changes in total income, ranging from a 0.7% decrease to a 0.6% increase. With fixed effects, these results become significant, indicating that the minimum wage increase modestly reduced total income. Specifically, the regression estimates a 0.7% additional decrease in total income for Nebraska relative to Oklahoma, which decreases to 0.6% after including demographic controls. While significant at the 10% level, the effect size is small, suggesting limited practical implications for young workers' total income.

These findings raise critical questions about the efficacy of minimum wage increases in achieving their intended goals, particularly for young workers in low-wage positions. The observed decreases in both hourly wages and total income suggest that employers may have adjusted other aspects of compensation to offset the increased labor costs. Potential mechanisms include reducing hours worked, altering the composition of jobs, or limiting wage growth for higher-paid workers. While the fixed-effects models provide robust evidence of the treatment effect, the modest magnitude of these effects highlights the complexity of labor market dynamics and the potential for unintended consequences.

In conclusion, the 2015 minimum wage increase in Nebraska had a statistically significant but modest impact on young workers' wages and total income. The results underscore

the importance of including state and time fixed effects in DiD analyses to address unobserved heterogeneity and omitted variable bias. Future research should explore the mechanisms driving these outcomes, such as changes in employment composition, hours worked, and other forms of compensation. Additionally, further investigation into the interaction between fixed effects and multicollinearity would provide valuable insights for policymakers seeking to evaluate labor market interventions.

Figure 1: Balance Test Regression

VARIABLES	(1) Balance Test Regression
ln_inctot (Natural log of Total Income)	0.00537 (0.0107)
ln_school	-0.107*** (0.00775)
ln_incwage (Natural Log of Wage Income)	0.00752 (0.0107)
ln_hrly_wage (Natural Log of Hourly Wage)	-0.0177*** (0.00558)
race_black	0.199*** (0.0150)
race_native_american	0.342*** (0.0140)
race_chinese	0.0437 (0.0687)
race_japanese	-0.195 (0.190)
race_other_asian_pacific	0.0801*** (0.0303)
race_other	0.170*** (0.0220)
race_two_major_races	0.310*** (0.0134)
race_three_or_more_races	0.265*** (0.0606)
female	-0.0101 (0.00674)
Constant	0.552*** (0.0288)
Observations	19,322
R-squared	0.078
Standard errors in parentheses	F(13, 19308) = 124.91
*** p<0.01, ** p<0.05, * p<0.1	Prob > F = 0.0000

Figure 2: Main Outcome Variables

Table of Means				
VARIABLES	(1) Oklahoma Before	(2) Oklahoma After	(3) Nebraska Before	(4) Nebraska After
ln_inctot	8.703 (0.0155)	8.841 (0.0141)	8.904 (0.0119)	9.034 (0.0107)
ln_hrly_wage	2.089 (0.00854)	2.227 (0.00777)	2.098 (0.00660)	2.213 (0.00600)
ln_incwage	8.642 (0.0157)	8.796 (0.0142)	8.844 (0.0122)	8.985 (0.0109)
in_school	0.608 (0.00575)	0.604 (0.00514)	0.463 (0.00453)	0.470 (0.00406)
race_black	0.0337 (0.00212)	0.0338 (0.00190)	0.0652 (0.00224)	0.0608 (0.00195)
race_native_american	0.0123 (0.00130)	0.0146 (0.00126)	0.0910 (0.00261)	0.0975 (0.00242)
race_chinese	0.00263 (0.000603)	0.00386 (0.000652)	0.00223 (0.000429)	0.00391 (0.000508)
race_japanese	0.000554 (0.000277)	0.000883 (0.000312)	0.000165 (0.000117)	0.000597 (0.000199)
race_other_asian_pacific	0.0121 (0.00128)	0.0192 (0.00144)	0.0126 (0.00102)	0.0175 (0.00107)
race_other	0.0157 (0.00146)	0.0222 (0.00155)	0.0288 (0.00152)	0.0325 (0.00144)
race_two_major_races	0.0216 (0.00171)	0.0295 (0.00178)	0.0961 (0.00268)	0.0951 (0.00239)
race_three_or_more_races	0.00139 (0.000438)	0.00353 (0.000623)	0.00405 (0.000577)	0.00709 (0.000683)
female	0.499 (0.00589)	0.490 (0.00525)	0.478 (0.00454)	0.487 (0.00407)
Observations	7,218	9,060	12,104	15,083

Oklahoma is the Treatment State
Standard errors in parentheses

Figure 3.1: Regression Estimations of Effect of Minimum Wage Increase on Wages

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Ln(HrlyWage)	Ln(HrlyWage)	Ln(HrlyWage)	Ln(HrlyWage)	Ln(HrlyWage)	Ln(HrlyWage)	Ln(HrlyWage)	Ln(HrlyWage)
treatment	0.008 (0.011)	-0.028*** (0.011)	0.016 (0.011)	-0.019* (0.011)				
post	0.138*** (0.012)	0.137*** (0.011)	0.138*** (0.012)	0.137*** (0.011)	0.185** (0.010)	0.185** (0.013)	0.138*** (0.001)	0.137*** (0.000)
post_treatment	-0.023 (0.015)	-0.020 (0.014)	-0.022 (0.015)	-0.020 (0.014)	-0.023*** (0.000)	-0.020*** (0.000)	-0.022*** (0.000)	-0.020*** (0.000)
yeardummy1					-0.084 (0.015)	-0.072 (0.013)		
yeardummy2					-0.045* (0.005)	-0.045* (0.004)		
yeardummy3					-0.021* (0.003)	-0.020** (0.000)		
o.yeardummy4					-	-		
yeardummy5					-0.133 (0.027)	-0.132 (0.029)		
yeardummy6					-0.113 (0.035)	-0.109 (0.036)		
yeardummy7					-0.079** (0.005)	-0.079** (0.002)		
yeardummy8					-0.097 (0.018)	-0.092* (0.013)		
o.yeardummy9					-	-		
in_school		-0.253*** (0.007)		-0.257*** (0.007)		-0.252** (0.015)		-0.257** (0.013)
race_native_american			-0.060*** (0.014)	-0.084*** (0.014)			-0.060** (0.004)	-0.084** (0.002)
race_other_asian_pacific			0.142*** (0.036)	0.164*** (0.036)			0.142 (0.100)	0.164 (0.099)
race_other			-0.006 (0.022)	-0.034 (0.022)			-0.006 (0.024)	-0.034 (0.031)
race_two_major_races			-0.034** (0.014)	-0.036*** (0.014)			-0.034 (0.012)	-0.036 (0.012)
race_three_or_more_races			-0.077 (0.054)	-0.078 (0.054)			-0.077 (0.045)	-0.078 (0.046)
Constant	2.089*** (0.009)	2.243*** (0.009)	2.089*** (0.009)	2.246*** (0.009)	2.132*** (0.002)	2.259*** (0.005)	2.099*** (0.001)	2.234*** (0.008)
Observations	43,471	43,471	43,471	43,471	43,471	43,471	43,471	43,471
R-squared	0.007	0.036	0.008	0.038	0.010	0.039	0.008	0.038
Robust standard errors in parentheses								
*** p<0.01, ** p<0.05, * p<0.1								

Figure 3.2: Regression Estimations of Effect of Minimum Wage Increase on Total Income

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
VARIABLES	Ln(IncTot)	Ln(IncTot)	Ln(IncTot)	Ln(IncTot)	Ln(IncTot)	Ln(IncTot)	Ln(IncTot)	Ln(IncTot)
treatment	0.201*** (0.020)	0.014 (0.017)	0.207*** (0.020)	0.032* (0.017)				
post	0.138*** (0.021)	0.134*** (0.018)	0.138*** (0.021)	0.134*** (0.018)	0.171*** (0.000)	0.172* (0.014)	0.138*** (0.002)	0.134*** (0.001)
post_treatment	-0.007 (0.026)	0.006 (0.023)	-0.007 (0.026)	0.006 (0.023)	-0.007*** (0.000)	0.006* (0.001)	-0.007** (0.000)	0.006* (0.001)
yeardummy1					-0.177 (0.034)	-0.120 (0.027)		
yeardummy2					-0.055 (0.024)	-0.055 (0.017)		
yeardummy3					-0.016 (0.028)	-0.011 (0.014)		
o.yeardummy4					-	-		
yeardummy5					-0.132 (0.021)	-0.125 (0.032)		
yeardummy6					-0.131 (0.068)	-0.113 (0.071)		
yeardummy7					-0.107 (0.036)	-0.107 (0.020)		
yeardummy8					-0.106 (0.031)	-0.077** (0.005)		
o.yeardummy9					-	-		
in_school		-1.286*** (0.011)		-1.289*** (0.011)		-1.284* (0.109)		-1.289* (0.108)
race_native_american			-0.028 (0.026)	-0.148*** (0.024)			-0.028 (0.018)	-0.148 (0.051)
race_other_asian_pacific			-0.019 (0.052)	0.090* (0.047)			-0.019 (0.153)	0.090 (0.144)
race_other			0.162*** (0.038)	0.018 (0.035)			0.162 (0.057)	0.018 (0.027)
race_two_major_races			-0.078*** (0.025)	-0.091*** (0.023)			-0.078* (0.006)	-0.091** (0.006)
race_three_or_more_races			-0.230** (0.095)	-0.232** (0.093)			-0.230** (0.013)	-0.232** (0.007)
Constant	8.703*** (0.015)	9.484*** (0.015)	8.703*** (0.016)	9.489*** (0.015)	8.891*** (0.004)	9.539*** (0.057)	8.832*** (0.005)	9.509*** (0.062)
Observations	43,465	43,465	43,465	43,465	43,465	43,465	43,465	43,465
R-squared	0.008	0.238	0.008	0.239	0.010	0.239	0.008	0.239
Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1								

Figure 4:

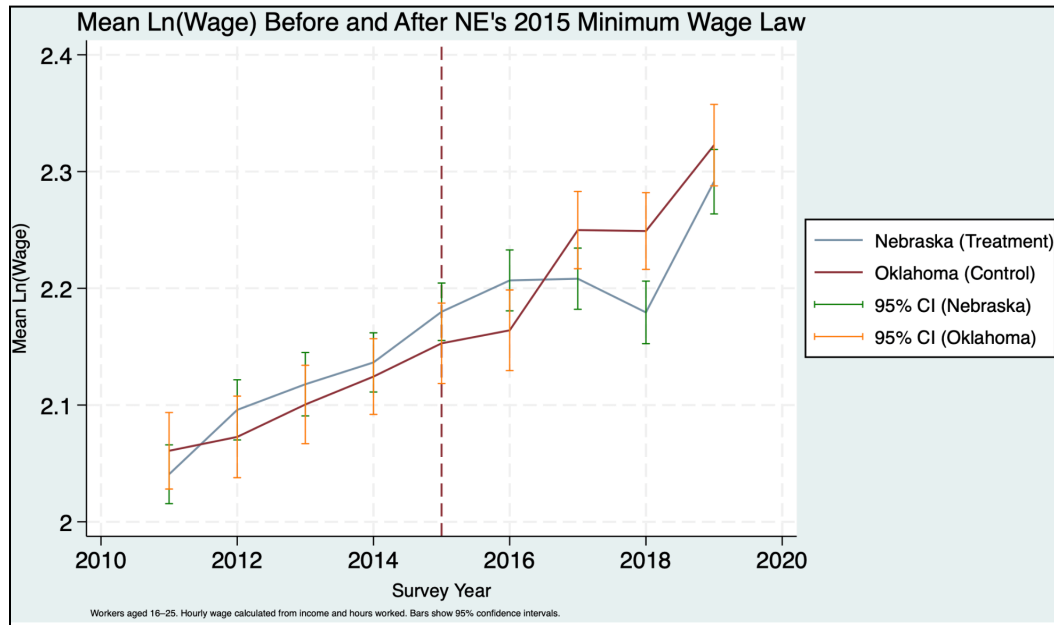


Figure 4 illustrates the trend of mean ln(wage) in Nebraska and Oklahoma from 2011 to 2019. The policy was implemented in 2015 as shown by the vertical red dashed line.

Figure 5:

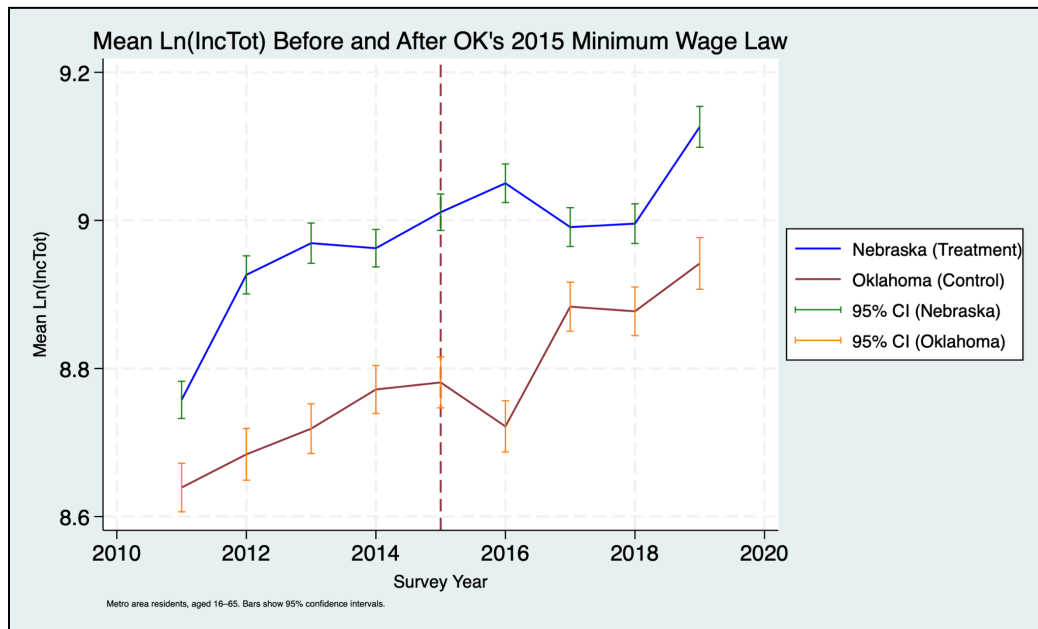


Figure 5 illustrates the trend of mean natural log of total income in Nebraska and Oklahoma from 2011 to 2019. The policy was implemented in 2015 as shown by the vertical red dashed line.

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